

CSE214 – Software Engineering Sessional
Offline on Performance and Load Testing

Report on LoadLab Performance

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YouTube Video Demonstration:
<https://youtu.be/7qkEvcth1Rk?si=FfufbaZWguLESZgM>

1 JMeter Setup

1.1 JMeter Test Plan

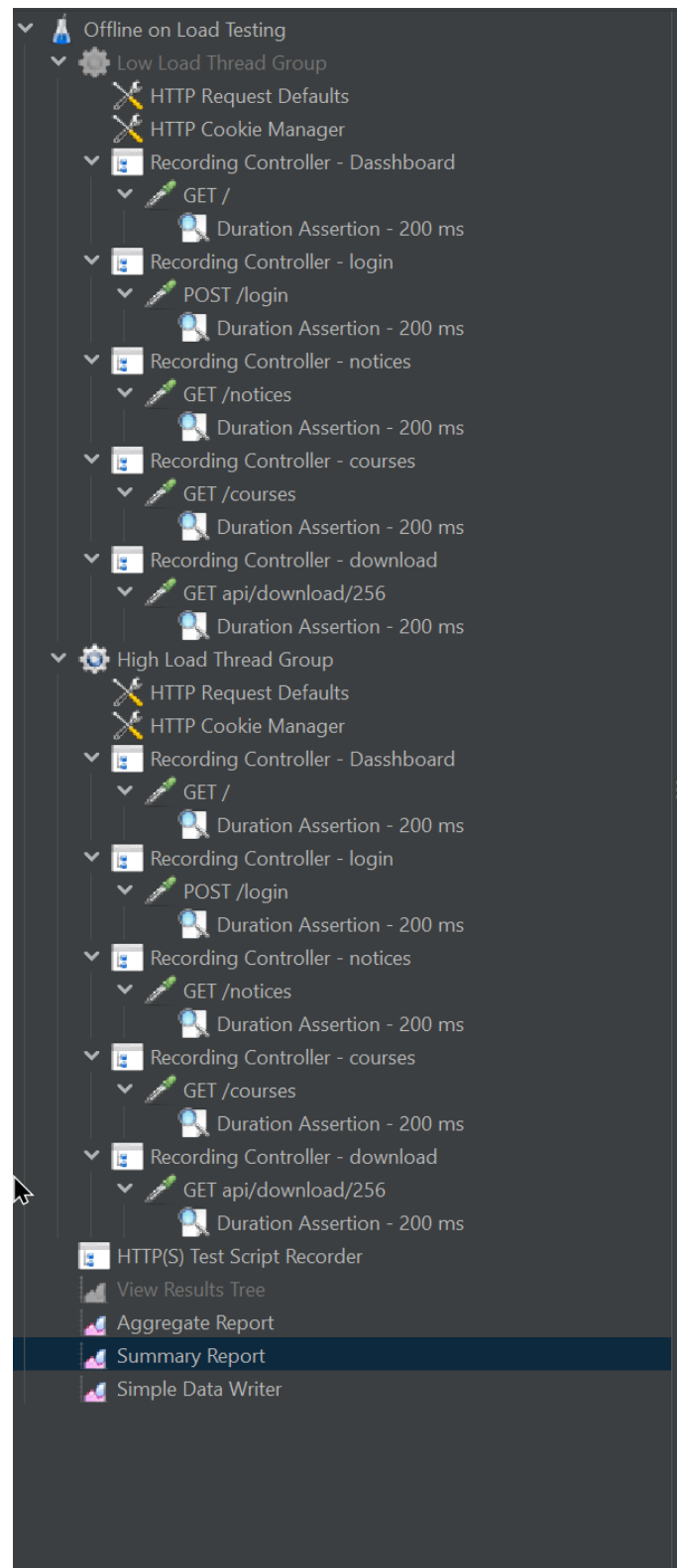


Figure 1: Final JMeter test-plan tree containing high- and low-load profiles, Recording Controllers, Assertions, listeners, and an HTTP(S) Test Script Recorder.

1.2 HTTP Request Defaults

The screenshot shows the 'HTTP Request Defaults' configuration window. The 'Name' field is set to 'HTTP Request Defaults'. The 'Basic' tab is selected. Under 'Web Server', the 'Protocol (http)' is 'http', 'Server Name or IP' is '103.94.135.91', and 'Port Number' is '8080'. Under 'HTTP Request', the 'Path' is '/' and 'Content encoding' is empty. The 'Parameters' tab is also visible, showing a table with columns: Name, Value, URL Encode?, Content-Type, and Include Equals?. At the bottom, there are buttons: Detail, Add, Add from Clipboard, Delete, Up, and Down.

Figure 2: HTTP Request Defaults configured for the LoadLab server.

1.3 Load Profiles

The screenshot shows the 'Thread Group' configuration window. The 'Name' field is set to 'Low Load Thread Group'. The 'Action to be taken after a Sampler error' is set to 'Continue'. Under 'Thread Properties', 'Number of Threads (users)' is 50, 'Ramp-up period (seconds)' is 100, and 'Loop Count' is 1. The 'Same user on each iteration' checkbox is checked. The 'Duration (seconds)' and 'Startup delay (seconds)' fields are empty.

(a) 50-thread profile.

The screenshot shows the 'Thread Group' configuration window. The 'Name' field is set to 'High Load Thread Group'. The 'Action to be taken after a Sampler error' is set to 'Continue'. Under 'Thread Properties', 'Number of Threads (users)' is 100, 'Ramp-up period (seconds)' is 100, and 'Loop Count' is 1. The 'Same user on each iteration' checkbox is checked. The 'Duration (seconds)' and 'Startup delay (seconds)' fields are empty.

(b) 100-thread profile.

Figure 3: Thread Group configurations used for the two workload profiles.

1.4 HTTP(S) Test Script Recorder

The JMeter HTTP(S) Test Script Recorder was configured on local proxy port 8888. Browser traffic was routed through the recorder via browser proxy settings.

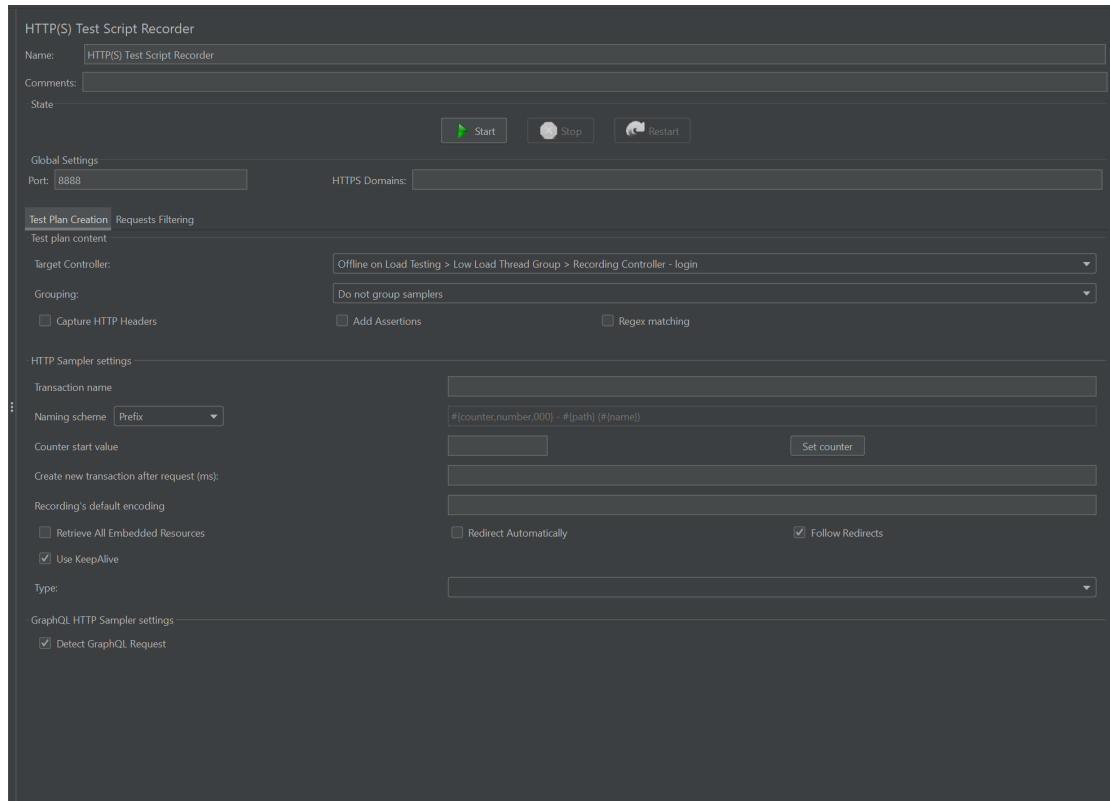


Figure 4: HTTP(S) Test Script Recorder configuration, including proxy port and target Recording Controller.

1.5 Authentication Request and Session Handling

The login sampler used the POST method with the path `/login`. The username and password were configured as request parameters. Follow Redirects was enabled, allowing JMeter to process the authentication redirect chain. An HTTP Cookie Manager maintained session cookies separately for each JMeter thread.

HTTP Request

Name: POST /login

Comments:

Basic Advanced

Web Server

Protocol (http): Server Name or IP: Port Number:

HTTP Request

POST Path: /login Content encoding:

☐ Redirect Automatically ☒ Follow Redirects ☒ Use KeepAlive ☐ Use multipart/form-data ☐ Browser-compatible headers

Parameters Body Data Files Upload

Send Parameters With the Request:

Name	Value	URL Encode?	Content-Type	Include Equals?
username	student	☐	text/plain	☒
password	student123	☐	text/plain	☒

Detail Add Add from Clipboard Delete Up Down

Figure 5: POST /login sampler showing the username and password parameters.

2 Duration Assertion

A Duration Assertion of

$$T = 200 \text{ ms}$$

was attached to each of the five main HTTP samplers. The assertion was configured for the main sample only.

The 200 ms value was suitable for this experiment because it produced both passing and failing requests.

Duration Assertion

Name: Duration Assertion - 200 ms

Comments:

Apply to:

☐ Main sample and sub-samples ☒ Main sample only ☐ Sub-samples only

Duration to Assert

Duration in milliseconds: 200

Figure 6: Duration Assertion configured with a maximum permitted response time of 200 ms.

An assertion failure does not necessarily indicate an HTTP transport or application failure. A request may return HTTP 200 successfully but still be classified by JMeter as failed if its elapsed time exceeds 200 ms.

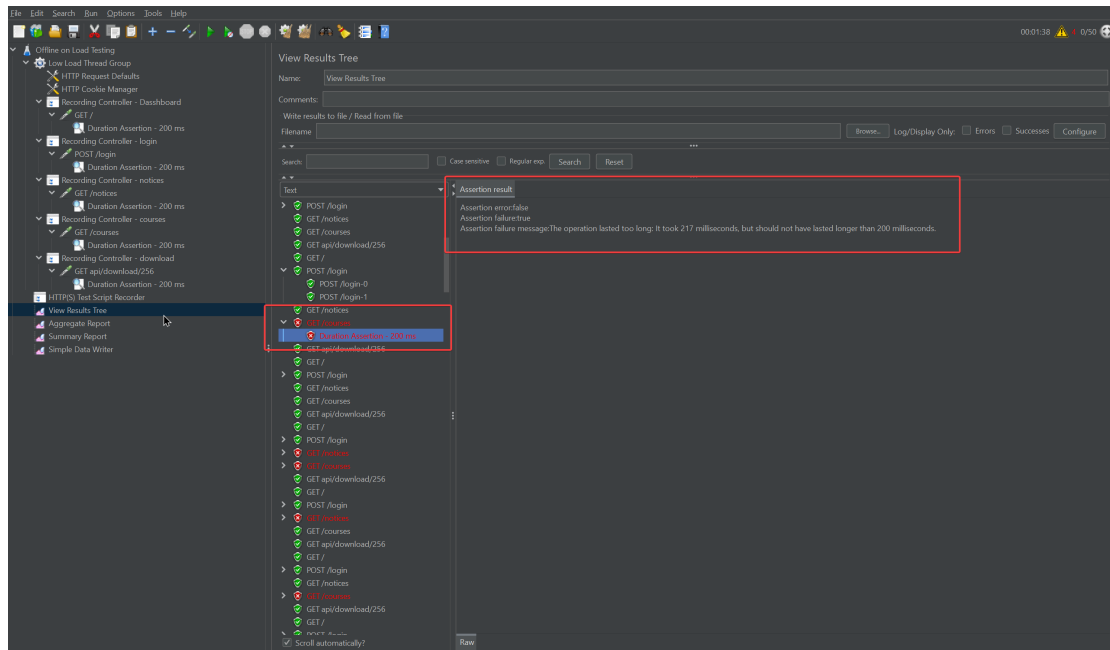


Figure 7: Example of a request that returned a response but failed the 200 ms Duration Assertion.

3 Experimental Results

3.1 Execution Time, Throughput, and Error Rate

Table 1 presents the required minimum, maximum, average, throughput, and error-rate values for each endpoint.

Table 1: Execution-time and throughput comparison

Endpoint	Threads	Min (ms)	Max (ms)	Avg (ms)	Throughput (req/s)	Error %
GET /	50	19	107	41	0.51007	0
GET /	100	14	92	38	1.00983	0
POST /login	50	90	404	162	0.50967	14
POST /login	100	74	751	140	1.00896	5
GET /notices	50	114	412	266	0.50822	68
GET /notices	100	109	412	262	1.00871	72
GET /courses	50	178	460	243	0.50816	84
GET /courses	100	168	298	233	1.00826	81
GET /api/download/256	50	26	457	56	0.50901	4
GET /api/download/256	100	25	573	44	1.00958	1

3.2 Bandwidth Comparison

Table 2: Throughput and network-rate comparison

Endpoint	Threads	Throughput (req/s)	Received (KB/s)	Sent (KB/s)
GET /	50	0.51007	1.21	0.06
GET /	100	1.00983	2.40	0.12
POST /login	50	0.50967	76.83	0.26
POST /login	100	1.00896	152.09	0.52
GET /notices	50	0.50822	3.79	0.14
GET /notices	100	1.00871	7.52	0.27
GET /courses	50	0.50816	22.10	0.14
GET /courses	100	1.00826	43.84	0.27
GET /api/download/256	50	0.50901	130.45	0.14
GET /api/download/256	100	1.00958	258.73	0.28

The Aggregate Report totals were:

Table 3: Overall aggregate metrics

Metric	50 Threads	100 Threads	Change
Average response time	154 ms	143 ms	−7.14%
Total throughput	2.53011 req/s	5.00791 req/s	+97.93%
Received rate	232.94 KB/s	461.06 KB/s	+97.93%
Sent rate	0.73 KB/s	1.44 KB/s	+97.26%
Error rate	34.0%	31.8%	−2.2 percentage points

4 Performance Analysis

4.1 Average Response-Time Comparison

The change in average response time between the two profiles was computed using

$$\% \Delta t = \frac{t_{100} - t_{50}}{t_{50}} \times 100\%.$$

Table 4: Change in average response time

Endpoint	50 Threads	100 Threads	Change
GET /	41 ms	38 ms	−7.32%
POST /login	162 ms	140 ms	−13.58%
GET /notices	266 ms	262 ms	−1.50%
GET /courses	243 ms	233 ms	−4.12%
GET /api/download/256	56 ms	44 ms	−21.43%

The 100-thread run did not produce higher average response time in this particular experiment. All five average values decreased slightly.

The two profiles used the same 100-second ramp-up and only one loop per thread, so the requests were introduced gradually rather than maintaining all configured threads under sustained simultaneous load. Run-to-run variation and uncontrolled activity on the shared class server may also have influenced the measurements.

Despite the lower averages, some worst-case values increased. The maximum login response time increased from 404 ms to 751 ms, while the maximum download response time increased from 457 ms to 573 ms. This demonstrates that average latency alone does not fully describe tail behavior.

5 Assertion and Error-Rate Analysis

The Duration Assertion was central to the observed JMeter error percentages. Table 5 compares the endpoint-level failure rates.

Table 5: Duration-Assertion-related error-rate comparison

Endpoint	50 Threads	100 Threads	Difference
GET /	0%	0%	0 pp
POST /login	14%	5%	−9 pp
GET /notices	68%	72%	+4 pp
GET /courses	84%	81%	−3 pp
GET /api/download/256	4%	1%	−3 pp

A failed JMeter sample must not automatically be interpreted as an HTTP protocol failure. During inspection, samples were observed that returned HTTP 200 OK but failed because the Duration Assertion detected an elapsed time greater than 200 ms.

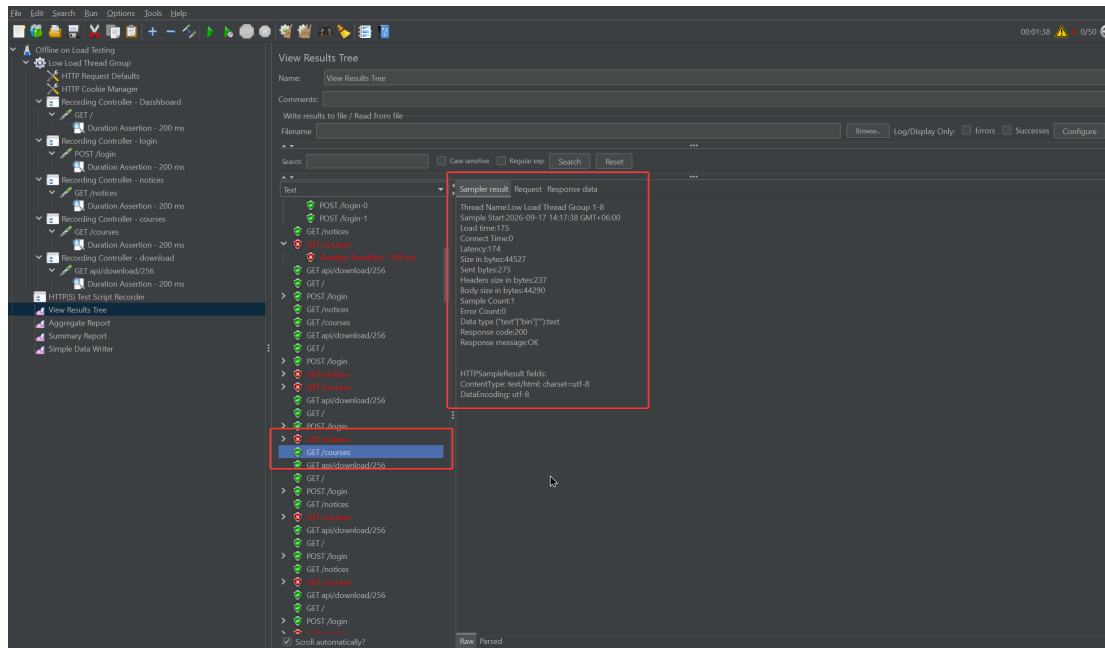


Figure 8: Example of a sample that completed within the assertion threshold.

6 HTML Dashboard Reports

JMeter HTML dashboards were generated from the raw result logs for both load profiles.

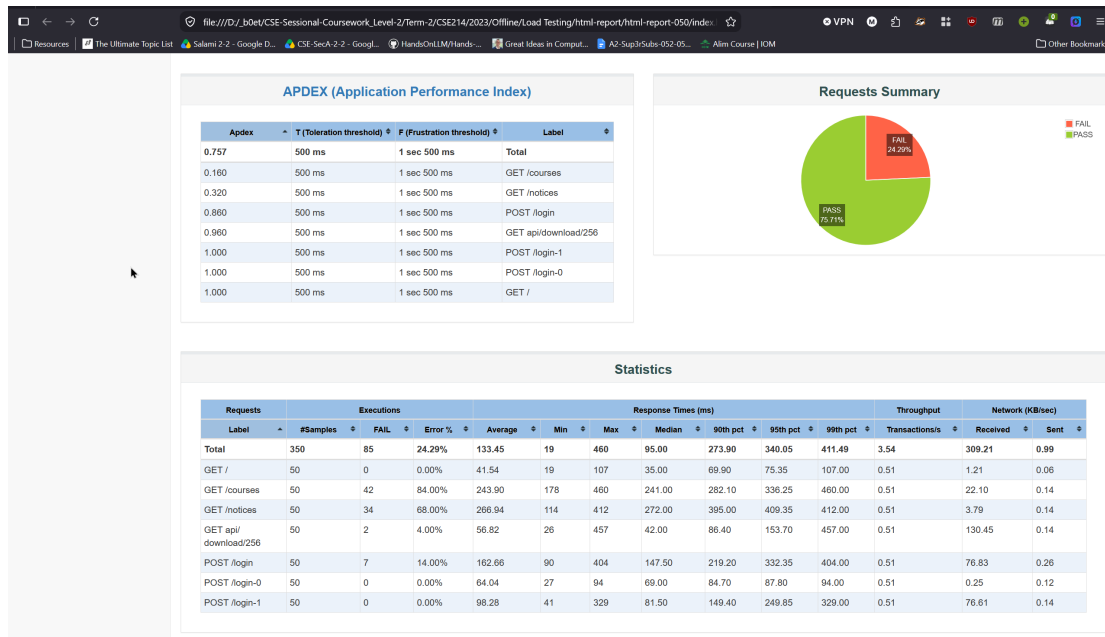


Figure 9: Generated JMeter HTML dashboard for the 50-thread test.

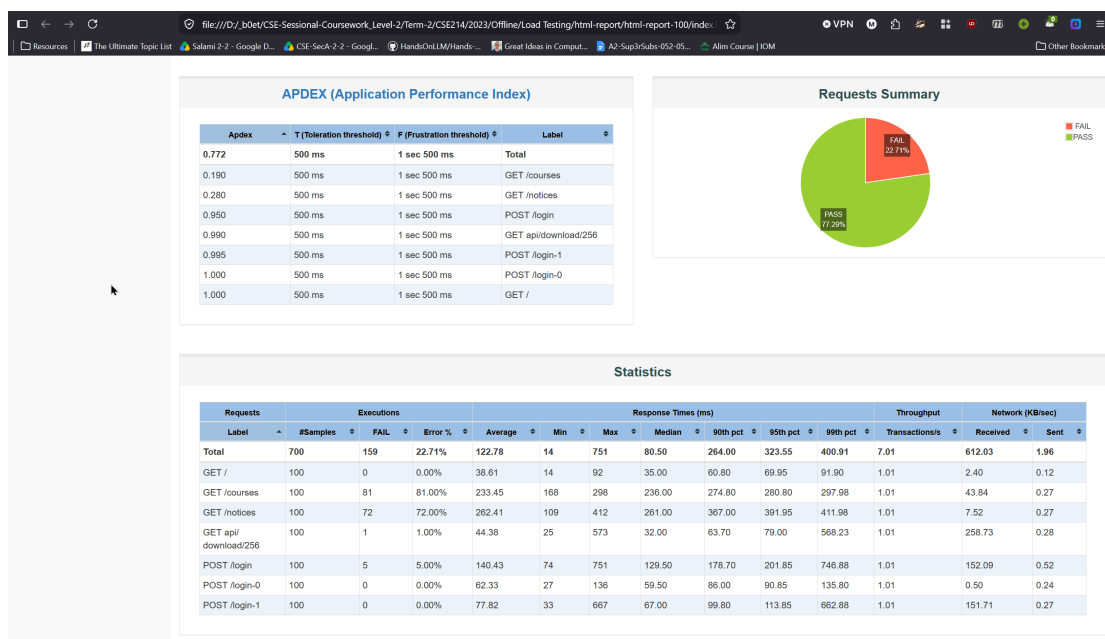


Figure 10: Generated JMeter HTML dashboard for the 100-thread test.